MODERNIZING FRAUD DETECTION FOR FEDERAL RESEARCH AWARDS

National Science Foundation | Office of Inspector General

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Fraud Indicator, SBIR/STTR awards, research compliance, dashboards

Summary:

Nicole modernized an existing compliance dashboard used to evaluate SBIR and STTR award adherence to NSF policy requirements. Her goal was to automate manual reporting processes, refresh the dashboard indicators, and redesign the user interface for clarity and ease of use. Using **SQL** to consolidate and clean award data, **Python** to automate recurring data pipeline tasks, and **PowerBI** to rebuild the visual reporting layer, Nicole delivered an updated dashboard that reduces manual upkeep and gives staff faster, more reliable visibility into compliance status across active awards.

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coding it forward >

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NICOLE LUCAS

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FEDERAL RESEARCH GRANTS

Project Overview

FEDERAL RESEARCH DOLLARS

- **SBIR** (Small Business Innovation Research) and **STTR** (Small Business Technology Transfer) programs award federal grants that fund early-stage R&D for startups.

Eligibility	Award Limits	Subcontracting

OIG's Office of Investigations is responsible for catching cases where those rules are being broken.

FRAUD INDICATOR DASHBOARD

A tool built for OIG's Office of Investigations to surface patterns in SBIR/STTR award that may indicate non-compliance.

The problem:

- No automated pipeline
- Unclear design
- Lacked some prioritization

DATA SOURCES

Research.gov
ONLINE GRANTS MANAGEMENT FOR THE NSF COMMUNITY

SAM.GOV

Award Cash Management $ervice

TOOLS AND TECH STACK

Data Sourcing & Querying: Pulled and structured award records from NSF administrative data systems using DBeaver, SQL and Python

Analysis & Indicator Logic: Rules and logic that translate raw fields into compliance-relevant flags. Utilized APIs to categorize data fields.

Dashboard & Visualization: Designed the interface investigators use to sort, filter, and prioritize flags using PowerBI

Severity Scoring: Layered a scoring approach to help rank what needs attention first

AUTOMATION PIPELINE

Data Lakehouse

DESIGN CONSIDERATIONS

- **Priority –** Indicator with weighted score to determine hierarchy in prioritization
- **Clear layout–** Minimal design that keeps relevant details front and center
- **Detailed Panels –** Drill through the dashboard for details on an institution and award

WHAT MADE THIS HARD

- **Ambiguous Signals**
  - Data alone can't tell the difference
- **Accessing sensitive data**
  - Constraints on data access due to privacy concerns

NEXT STEPS

Reviewed status– Investigators are aware of which institution has yet to be reviewed

Notification feature – Easily identify which institutions need to be reviewed with a certain priority

REFLECTIONS

- **Sharper questions** – asking what data means in practice, not just what a query returns
- **Context as a discipline** – treating surrounding facts as part of the analysis
- **Translation as a skill** – learning that how a finding is presented shaped whether it gets acted on

THANK YOU!

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